Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics

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BSE Summer Forum

Heterogeneous agent models with aggregate risk

- Huge literature since Krusell-Smith and Den Haan from late 90s

- Key challenge: rational expectations + general equilibrium
  $\Rightarrow$ distribution = state variable in Bellman equation ("Master equation")
  - true even though households/firms only care about prices
  - intuition: equilibrium prices are not Markov, only the distribution is
    $\Rightarrow$ forecast distributions to forecast prices

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  - intuition: equilibrium prices are not Markov, only the distribution is
    $\Rightarrow$ forecast distributions to forecast prices
- Despite recent impressive advances to solve it directly, still lack of efficient global solution methods for advanced HA models with aggregate risk
- This paper: sidestep master eqn with structural reinforcement learning

Sidestep Master eqn using structural reinforcement learning

RL: learn value & policy functions in Markov decision process with Monte Carlo.

Unlike dynamic programming, RL can handle non-Markov states (e.g. prices).

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This paper:

- Replace dist'n with low-dim. prices in state space
- RL about equilibrium prices but not individual states $\Rightarrow$ "Structural RL"
- Efficient asset market clearing using policy functions (= demand curves)
- Structural RL for both households and firms $\Rightarrow$ solving HANK

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Outcome: efficient & flexible global solution method for HA models w agg risk, solves problems traditional methods struggle with:

1. non-trivial market clearing (Huggett w agg. risk) $\approx$ 1 min on Google Colab
2. portfolio choice $\approx$ 1 min
3. HANK with forward-looking price/wage Phillips curve $\approx$ 4 min

Structural RL in a (math) nutshell

- Step 1: parameterize policy function in low-dim state (s, z, p):

$$\pi_\theta(s, z, p) = \{c_\theta(s, z, p), b'_\theta(s, z, p)\},$$

  with grid values as unknown parameters θ. Do <u>NOT</u> parameterize price functions.

- Step 2: GE simulation given π_θ. For any θ, compute average lifetime utility

$$v(\theta) \approx \frac{1}{N} \sum_{n=1}^{N} \sum_{t=0}^{T} \beta^t u(c_\theta(s_t^n, z_t^n, p_t^n))$$

  along N simulated equilibrium paths with very large T.

- Step 3: structural policy gradient. Update θ by stochastic gradient ascent on $v(\theta)$, and repeat Step 2.

Restricted perceptions equilibrium. Results so far very close to RE.

Literature and contribution

Global solution methods for HA models with aggregate risk

Fernández-Villaverde et al, Han-Yang-E, Maliar-Maliar-Winant, Azinovic-Gaegauf-Scheidegger, Schaab, Duarte-Silva...

- sidestep Master equation rather than "taming curse of dimensionality"
- Han-Yang-E DeepHAM = also RL-inspired

Global solution with bounded rationality Krusell-Smith, Den Haan, ...

- difference: no perceived law of motion

Self-confirming equilibrium & restricted perceptions equilibrium

- agents form price exp. from data generated by economy where they live
- but expectations incorrect for off-equilibrium (or rare) price realizations

Adaptive (least squares) learning Bray, Marcet-Sargent, Evans-Honkapohja, Jacobson, Giusto, ...

- similarity: stochastic approximation; difference: ours $\neq$ theory of learning

"Sequence space" Auclert- Bardóczy-Rognlie-Straub

- global solution in sequence space (low-dim. prices) via Monte Carlo

Plan

1. Textbook HA model with aggregate risk: setup and challenge
2. Structural RL for HA macro: sidestepping the Master equation
3. Computational experiments: Krusell-Smith and Huggett with agg risk
4. Forward looking Phillips curve: HANK with aggregate shocks.

Textbook HA model with aggregate risk

Textbook HA model – Huggett (1993) with aggregate risk

- Continuum of agents i, heterog. in $(b_{i,t}, y_{i,t})$, $y_{i,t}$ = id. risk, agg. shock z_t
- Households choose consumption $c_{i,t}$ to maximize

$$v_{i,0} = \max_{\{c_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{i,t}) \quad \text{subject to}$$

$$c_{i,t} + q_t b_{i,t+1} = b_{i,t} + y_{i,t} z_t, \quad y_{i,t+1} \sim \mathcal{T}_y(\cdot|y_{i,t}), \quad z_{t+1} \sim \mathcal{T}_z(\cdot|z_t), \; b_{i,t+1} \geq \underline{b}$$

- State of the economy: distribution $G_t(b, y)$ and agg. shock z_t
- Market clearing: bond price q_t such that

$$\int b'_t(b, y) \, dG_t(b, y) = 0, \quad \text{all } t$$

Note: agent problem depends on G_t only through low-dimensional price q_t

Discretized representation with general notation

- Discretize individual state $s = (b, y) \in \{s_1, ..., s_J\}$ with $J = J_b \times J_y$
- Value function, distribution, etc are J-dimensional vectors

$$\mathbf{v}_t = \begin{bmatrix} v_t(s_1) \\ \vdots \\ v_t(s_J) \end{bmatrix}, \qquad \mathbf{g}_t = \begin{bmatrix} g_t(s_1) \\ \vdots \\ g_t(s_J) \end{bmatrix}$$

- Consumption policy $c = \pi_t(s, z) \Rightarrow J \times J$ transition matrix for s

$$\mathbf{A}_{\pi_t(z_t)} \quad \text{with entries} \quad \Pr(s_{j'}|s_j) = \mathcal{T}_s(s_{j'}|s_j, \pi_t(s_j, z_t), p_t, z_t)$$

- High-dimensional state $(\mathbf{g}_t, z_t)$ is Markov:

$$\mathbf{g}_{t+1} = \mathbf{A}^{\top}_{\pi_t(z_t)} \mathbf{g}_t, \quad z_{t+1} \sim \mathcal{T}_z(\cdot|z_t)$$

- Low-dim equilibrium prices $p_t = P^*(\mathbf{g}_t, z_t)$: not Markov.
  Note: in Huggett, the low-dimensional price is $p_t = q_t$.

Key difficulty: equilibrium prices are not Markov

- Low-dim equilibrium prices $p_t = P^*(\mathbf{g}_t, z_t)$: not Markov
- ... only extremely high-dimensional $(\mathbf{g}_t, z_t)$ is
- Dynamic programming can only handle Markov states $\Rightarrow$ Master equation

$$V(s, \mathbf{g}, z) = \max_c u(c) + \beta \mathbb{E}\left[V(s', \mathbf{g}', z') | s, \mathbf{g}, z\right] \text{ s.t. } s' \sim \mathcal{T}_s(\cdot | s, c, P^*(\mathbf{g}, z))$$

- Without Markov transition prob's: cannot even write Bellman equation!
- Our solution: use RL to approximate value/policy functions with low-dim non-Markov state variables

Sidestepping the Master Equation via Structural RL

What is reinforcement learning? A simple example

RL = learning value & policy functions from Monte Carlo ▸ RL Primer

Example: compute value function (eliminate actions & individual states for now)

$$v(p) = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u(p_t) \middle| p_0 = p\right], \quad p_t = \text{exogenous stochastic process}$$

Two approaches:

1. Dynamic programming: p_t Markov and know $f(p'|p)$

$$v(p) = u(p) + \beta \int v(p') f(p'|p) dp'$$

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2. RL/Monte Carlo: don't know f but sample N trajectories $\{p_t^i\}_{t=0}^{T}$

$$v(p) \approx \hat{v}(p) = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \beta^t u(p_t^i) \quad \text{or} \quad \hat{v}^k = \hat{v}^{k-1} + \frac{1}{k}\left(\sum_{t=0}^{T} \beta^t u(p_t^k) - \hat{v}^{k-1}\right)$$

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Can also be extended to compute optimal policy: policy gradient method

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$$v_{i,0} = \max_{\{c_{i,t}\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u(c_{i,t})\right] \text{ s.t. } s_{i,t+1} \sim \mathcal{T}_s(\cdot | s_{i,t}, c_{i,t}, z_t, p_t), \quad p_t = P^*(G_t, z_t) \tag{1}$$

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Assumption 1: agents observe prices p_t but not distribution $G_t(s)$ ▸ Wold repr.

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Assumption 2: consumption policy π does not condition on price histories

$$c_{i,t} = \pi(s_{i,t}, p_t, z_t)$$

- Extension: keep track of price histories (h lags or RNN) $\Rightarrow$ similar results

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RL: optimize $\pi(s_{i,t}, z_t, p_t)$ to solve (1) on the simulated paths. ▸ RL Primer

Similarity to standard RL: don't know transition prob. of prices p_t

Difference to standard RL: know "local environment" $\mathcal{T}_s$ and u (Han-Yang-E)

Restricted perceptions equilibrium

A pair of mappings (π^*, P^*) constitutes a restricted perceptions equilibrium if:

1. Optimality. For any price sequence $\{p_t\}$ generated by $p_t = P^*(\mathbf{g}_t, z_t)$ and exogenous sequence $\{z_t\}$, agents choose $\pi^*(s, p, z)$ to solve:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u(\pi(s_t, z_t, p_t))\right],$$

   subject to the individual budget constraint and state transition equations.

2. Market clearing. For every period t, all markets clear.

3. Consistency. The prices that agents use to form expectations coincide with the prices in the simulated economy when all agents follow π^*:

$$p_t = P^*(\mathbf{g}_t, z_t), \qquad \mathbf{g}_{t+1} = \mathbf{A}^\top_{\pi^*(z_t, p_t)} \mathbf{g}_t$$

Simulating the economy given policy $c = \pi_\theta(s, z, p)$

Given policy $\pi_\theta(s, z, p)$ and $(\mathbf{g}_0, z_0)$, want to simulate economy forward:

$$\{\mathbf{g}_t, p_t, z_t\}_{t \geq 0}$$

- important: very cheap computationally with JAX on GPUs

Recall: discrete $s \Rightarrow$ vectors $\boldsymbol{\pi}(z, p)$, $\mathbf{g}_t$, sparse transition matrix $\mathbf{A}_{\pi(z,p)}$

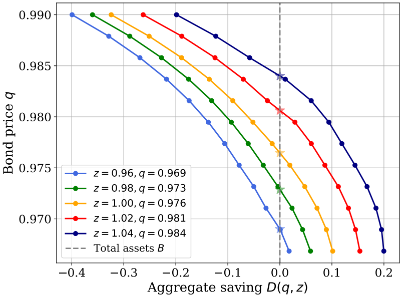
- Simulating $z_{t+1} \sim \mathcal{T}_z(\cdot|z_t)$: easy
- Simulating $\mathbf{g}_{t+1} = \mathbf{A}^\top_{\pi(z_t, p_t)} \mathbf{g}_t$ given p_t: easy
- Simulating $p_t = P^*(\mathbf{g}_t, z_t)$: hard! Pinned down implicitly by market clearing each period

Efficient handling of non-trivial market clearing

$$D_t(p, z) = 0, \qquad D_t(p, z) := \int b'(s, z, p) dG_t(s) = \text{agg. demand for bonds}$$

Key: policies $b'(s, z, p)$ double as individual demand functions

$$p_t \ \text{ solves } \ D_t(p_t, z_t) = \mathbf{b}'(z_t, p_t)^\top \mathbf{g}_t = 0$$



Market clearing is part of environment, not another loop!

Using structural knowledge of individual dynamics

Value function for given policy $\pi(s, z, p) = \{c(s, z, p), b'(s, z, p)\}$

$$v_\pi(s, z, p) = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u(\pi(s_{i,t}, z_t, p_t)) \middle| s_{i,0} = s, z_0 = z, p_0 = p\right] \qquad (*)$$

Partition state space (s, z, p) into known dynamics and unknown dynamics

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Partition state space (s, z, p) into known dynamics and unknown dynamics

- use transition matrix $\mathbf{A}$ to keep track of all s-transitions
- expectation $\mathbb{E}$ only over price trajectories $\{p_t\}_{t=0}^{\infty}$ and $\{z_t\}_{t=0}^{\infty}$

Write $v_\pi(s,z,p)$ in $(*)$ as vector $\mathbf{v}_\pi(z,p)$:

$$\mathbf{v}_\pi(z,p) = \mathbb{E}\left[\mathbf{u}_0 + \beta \mathbf{A}_0 \mathbf{u}_1 + \beta^2 \mathbf{A}_0 \mathbf{A}_1 \mathbf{u}_2 + \ldots | z, p\right] = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \mathbf{A}_{0\to t} \mathbf{u}_t \middle| z, p\right]$$

where $\mathbf{u}_t = u(\boldsymbol{\pi}(z_t, p_t))$ and $\mathbf{A}_t = \mathbf{A}_{\pi(z_t, p_t)}$ and $\mathbf{A}_{0\to t} = \mathbf{A}_0 \cdots \mathbf{A}_{t-1}$

Summary: problem to be solved

Find optimal policy $\pi(s, z, p)$ or $\boldsymbol{\pi}(z, p)$ that maximizes

$$\mathbf{v}_\pi(z,p) = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \mathbf{A}_{\pi,0\to t} u(\boldsymbol{\pi}(z_t, p_t)) \middle| z_0 = z, p_0 = p\right]$$

taking as given evolution of equilibrium prices p_t (sg[$\cdot$] = stop-gradient operator)

$$p_t = \mathrm{sg}\left[P^*(\mathbf{g}_t, z_t)\right], \quad \mathbf{g}_{t+1} = \mathbf{A}^{\top}_{\pi(z_t,p_t)} \mathbf{g}_t, \quad z_{t+1} \sim \mathcal{T}_z(\cdot|z_t), \quad t = 0, 1, \ldots$$

with $(\mathbf{g}_0, z_0)$ given and $\mathbf{A}_{\pi,0\to t} = \mathbf{A}_{\pi(z_0,p_0)} \cdots \mathbf{A}_{\pi(z_{t-1},p_{t-1})}$

Key observation:

- State of economy = $(\mathbf{g}, z)$ = very high-dimensional
- But state in value/policy functions = (s, z, p) = very low-dimensional!
- No perceived law of motion, inner loop / outer loop (like in Krusell-Smith)
- GE problem only mildly more difficult than PE

Structural policy gradient (SPG) method for maximizing $\mathbf{v}_\pi$

Find optimal policy $\boldsymbol{\pi}(z, p)$ that maximizes estimate of $\mathbb{E}_{z_0 \sim \psi_z, p_0 \sim \psi_p}[\mathbf{v}_\pi(z_0, p_0)]$:

$$\widehat{\mathbf{v}}_\pi = \frac{1}{N} \sum_{n=1}^{N} \left[\sum_{t=0}^{T} \beta^t \mathbf{A}^n_{\pi,0\to t} u(\boldsymbol{c}(z^n_t, p^n_t)) \right]$$

with N price trajectories p^n_t sample from interacting with environment (rollout):

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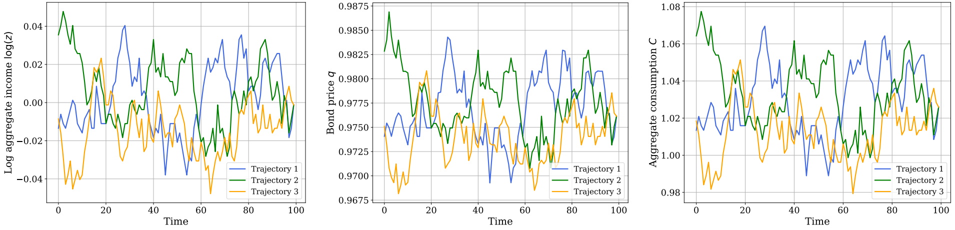

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with $\mathbf{g}^n_0 \sim \psi_g(\cdot)$, $z^n_0 \sim \psi_z(\cdot)$ and $\mathbf{A}^n_{\pi,0\to t} = \mathbf{A}_{\pi(z^n_0, p^n_0)} \cdots \mathbf{A}_{\pi(z^n_{t-1}, p^n_{t-1})}$

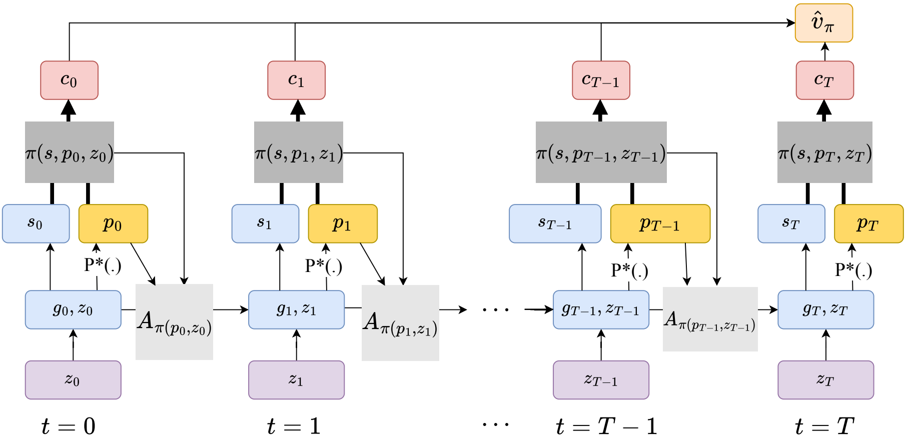
In practice, maximize scalar objective using gradient ascent:

$$\mathcal{L}(\boldsymbol{\theta}) = \mathbf{d}_0^\top \widehat{\mathbf{v}}_\pi = \sum_{j=1}^{J} d_0(s_j) \widehat{v}_\pi(s_j), \quad d_0(s) = \text{uniform dist. over } s$$

Policy either grid based $\boldsymbol{\theta} = [\boldsymbol{\pi}(z_1, p_1), \ldots, \boldsymbol{\pi}(z_K, p_L)]$ or neural net $\pi(s, z, p; \boldsymbol{\theta})$

- low dim. $\Rightarrow$ grid-based method works well, no need for neural nets!

Computational graph for construction of $\widehat{\mathbf{v}}_\pi$



Computational experiments

Runtimes

- Efficient implementation in JAX for GPUs, run on Google Colab
- Stochastic algorithm: present averages over multiple runs

Model	Average converge epoch	# Runs	Average Runtime (sec)
Krusell-Smith	462.3	10	36.77
Huggett with agg. shocks	573.0	10	45.91
Portfolio choice	637.5	10	84.3
HANK with agg. shocks	707.5	10	246.49
Partial Equilibrium (Huggett)	355.8	10	24.50

Note: all experiments were implemented on the A100 GPU on Google Colab

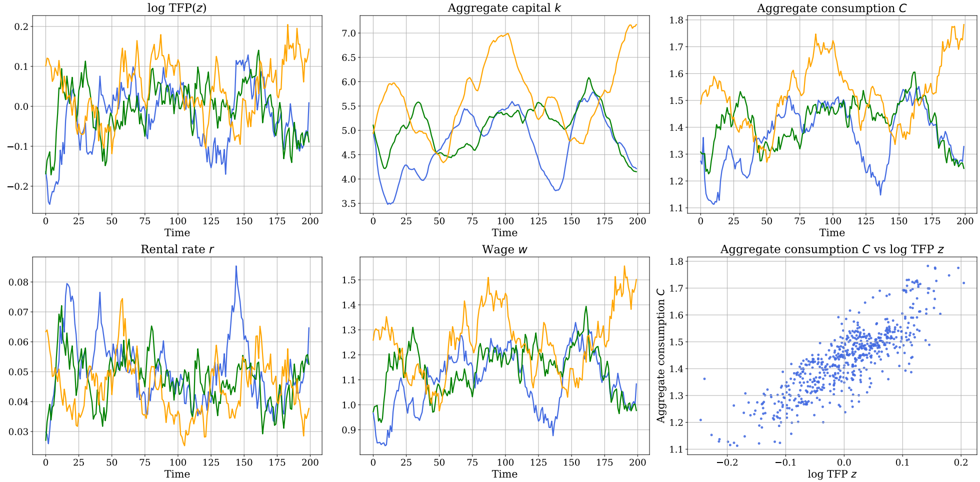
Krusell-Smith model

Computational experiments: Krusell-Smith model

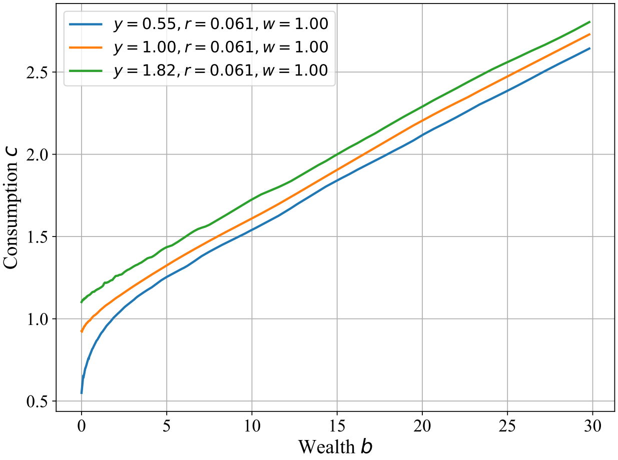
Parameter	Description	Value
α	Capital share	0.36
δ	Capital depreciation rate	0.08
γ	Discount factor	0.95
σ	Coefficient of relative risk aversion	3
ρ_z	Persistence of AR(1) for z_t (log TFP)	0.9
ν_z	Volatility of AR(1) for z_t (log TFP)	0.03

Hyperparameter	Description	Value
J_{s_1}	Number of s_1 (wealth) grid points	200
J_{s_2}	Number of s_2 (income) states	3
J_{p_1}	Number of p_1 (rental rate) grid points	50
J_{p_2}	Number of p_2 (wage) grid points	70
N	Sample size = number of p trajectories	32,128,512,...
T	Time truncation s.t. $\beta^T < 0.001$	135
ϵ	Convergence criterion on $\widehat{\mathbf{v}}_\pi$	1×10^{-4}
$\eta_{\mathtt{ini}}$	Initial learning rate	5×10^{-4}
$\eta_{\mathtt{decay}}$	Learning rate decay (exponential)	0.5

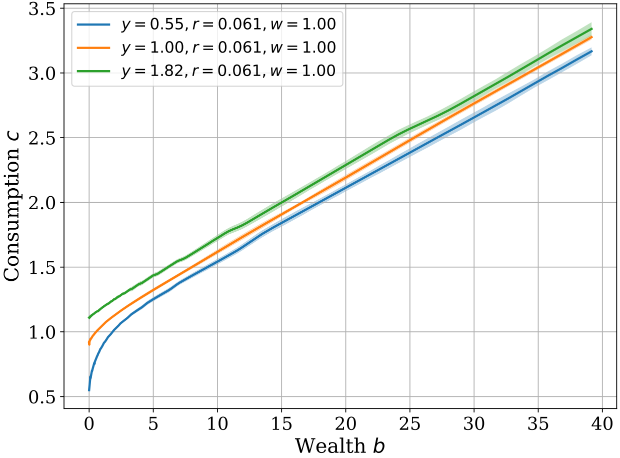
Some simulated trajectories under the optimal policy



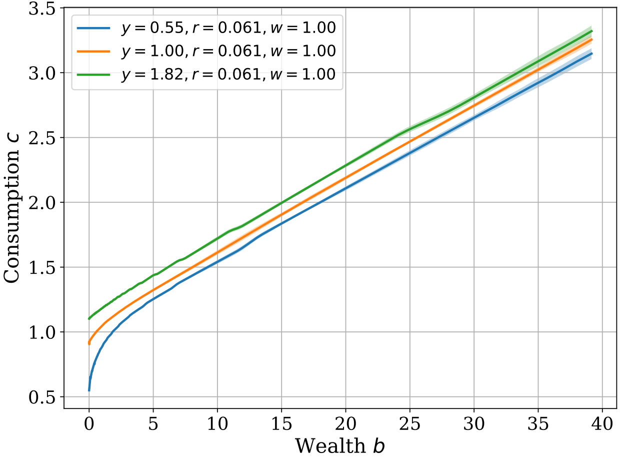
Consumption policy function: single run with $N_{\text{sample}} = 128$



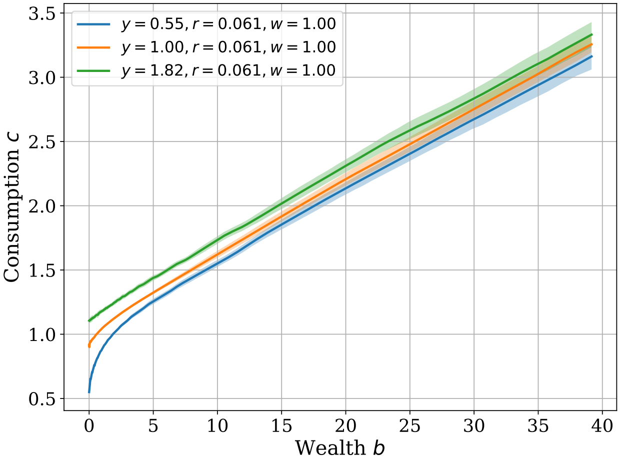
Consumption policy: multiple runs with $N_{\text{sample}} = 128$



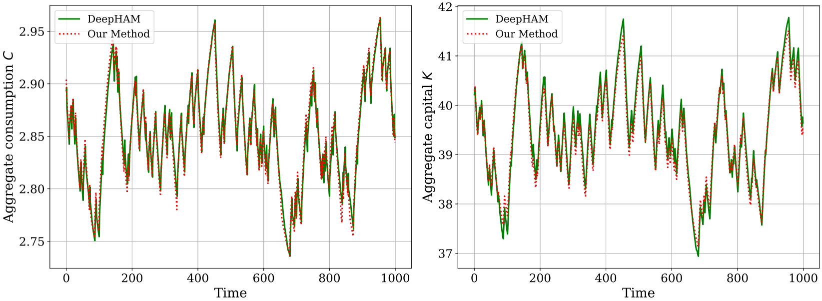
Larger sample size $N = 512 \Rightarrow$ more precise estimate



Smaller sample size $N = 32 \Rightarrow$ noisier estimate



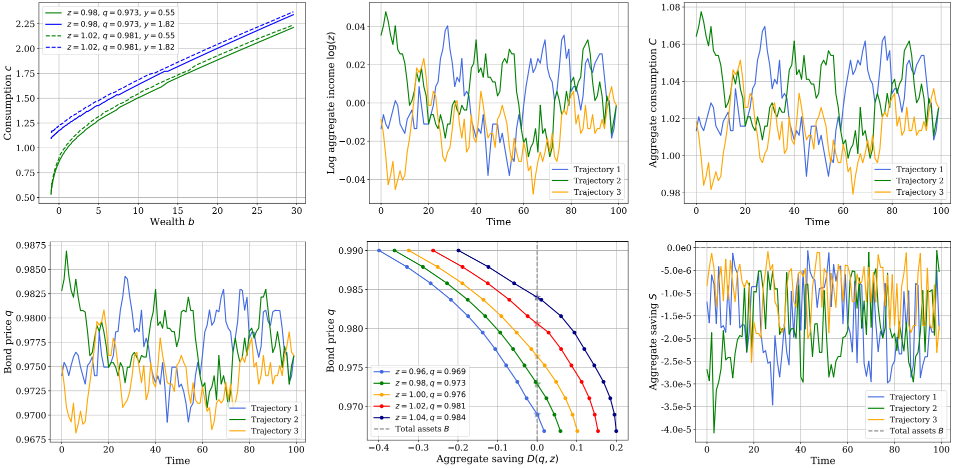
Structural RL method recovers RE solutions



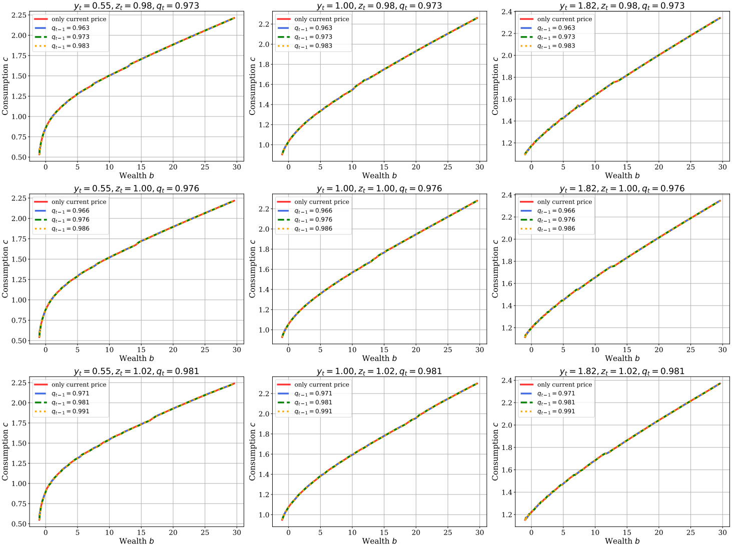
RE solutions are obtained with a deep learning based method (DeepHAM).

Huggett with aggregate risk
(Non-trivial market clearing)

Some simulated trajectories under the optimal policy



Adding one lagged price p_{t-1} into state space



Portfolio Choice

Portfolio choice: Krusell-Smith 1997

- RBC model with household portfolio choice
- Three individual states $(b_{i,t}, k_{i,t}, y_{i,t})$, three prices (q_t, R_t, w_t)
- Households choose consumption $c_{i,t}$ to maximize

$$v_{i,0} = \max_{\{c_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{i,t}) \quad \text{subject to}$$

$$c_{i,t} + q_t b_{i,t+1} + k_{i,t+1} = b_{i,t} + R_t k_{i,t} + w_t y_{i,t},\; y_{i,t+1} \sim \mathcal{T}_y(\cdot|y_{i,t}),\; b_{i,t+1} \geq \underline{b}$$

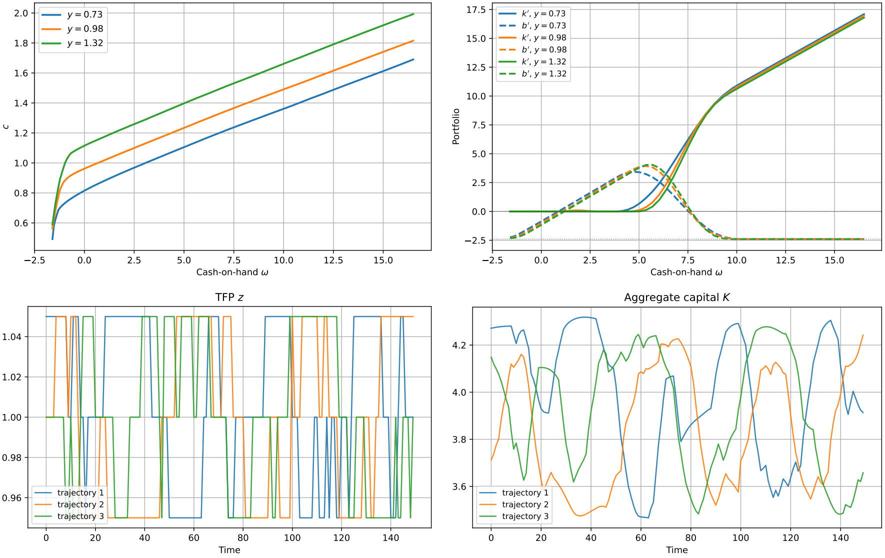
- Rental rate and wage

$$R_t = \alpha z_t K_t^{\alpha-1} + 1 - \delta, \quad w_t = (1-\alpha) z_t K_t^{\alpha}, \quad K_t = \int k \; dG_t(b, k, y)$$

- Bond price q_t such that

$$\int b_t'(b, k, y)\; dG_t(b, k, y) = 0, \quad \text{all } t$$

Some simulated trajectories under the optimal policy



HANK with forward-looking Phillips curve

HANK with forward-looking Phillips curve

Household block (similar to before): states $s = (b, y)$ and prices $p = (q, w)$

$$\text{policies} = \{c(s,p), n(s,p)\} \quad \text{that maximize PDV of utility}$$

Firm block: price setting $\Rightarrow$ forward-looking Phillips curve = added difficulty

$$\Pi_t = \frac{\varepsilon}{\theta}\left(\frac{w_t}{z_t} - m^*\right) + \mathbb{E}\left[\Lambda_{t\to t+1}\frac{Y_{t+1}}{Y_t}\Pi_{t+1}\,\middle|\,\mathcal{I}_t\right], \qquad m^* = \frac{\varepsilon - 1}{\varepsilon}$$

Conventional approach: parameterize $\mathbb{E}[\Pi_{t+1}|\mathcal{I}_t]$ $\Rightarrow$ complicated fixed-point
(e.g. Kase-Melosi-Rottner)

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Our solution: solve firm price-setting problem using policy gradient method

$$J_{j,0} = \max_{\{P_{j,t}\}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \Lambda_{0\to t} \left\{ \text{Profits}\left(\frac{P_{j,t}}{P_t}, \frac{w_t}{z_t}, Y_t \right) - \frac{\theta}{2} \left(\frac{P_{j,t} - P_{j,t-1}}{P_{j,t-1}} \right)^2 \right\} \right]$$

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Or in terms of relative price $\phi_{j,t} = P_{j,t}/P_t$ and inflation $\Pi_{j,t} = (P_{j,t} - P_{j,t-1})/P_{j,t}$

$$J_{j,0} = \max_{\{\Pi_{j,t}\}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \Lambda_{0\to t} \left\{ \text{Profits}\left(\phi_{j,t}, \frac{w_t}{z_t}, Y_t\right) - \frac{\theta}{2} \left(\Pi_{j,t}\right)^2 \right\} \right], \quad \phi_{j,t} = \frac{1+\Pi_{j,t}}{1+\Pi_t} \phi_{j,t-1}$$

HANK: policy gradient method for both households & firms

Household block (similar to before): states $s = (b, y)$ and prices $p = (q, w)$

$$\text{policies} = \{c(s,p), n(s,p)\} \quad \text{that maximize PDV of utility}$$

Firm block: states (z, Y) and prices $p = (q, w)$

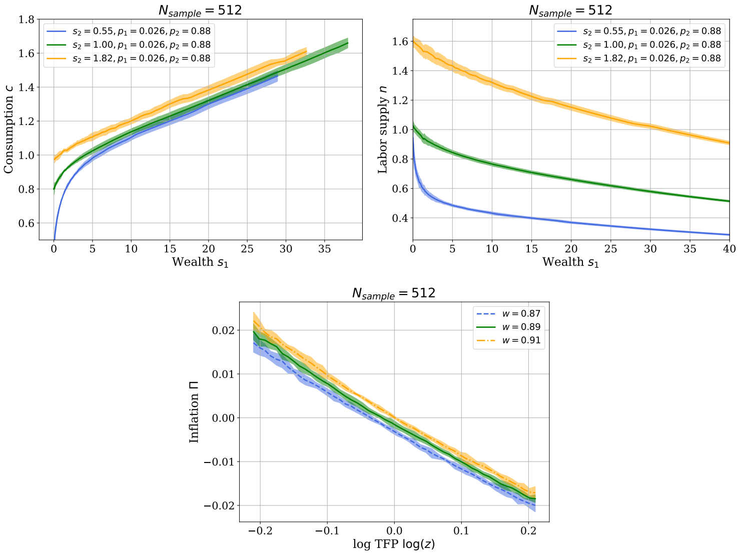
$$\text{policy} = \Pi_j(z, Y, p) \quad \text{that maximizes}$$

$$J_{j,\Pi} = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \Lambda_{0\to t} \left\{ \text{Profits}\left(\phi_{j,t}, \frac{w_t}{z_t}, Y_t\right) - \frac{\theta}{2}\left(\Pi_j(z_t, Y_t, p_t)\right)^2 \right\} \right], \phi_{j,t} = \frac{1 + \Pi_j(z_t, Y_t, p_t)}{1 + \Pi_t} \phi_{j,t-1}$$

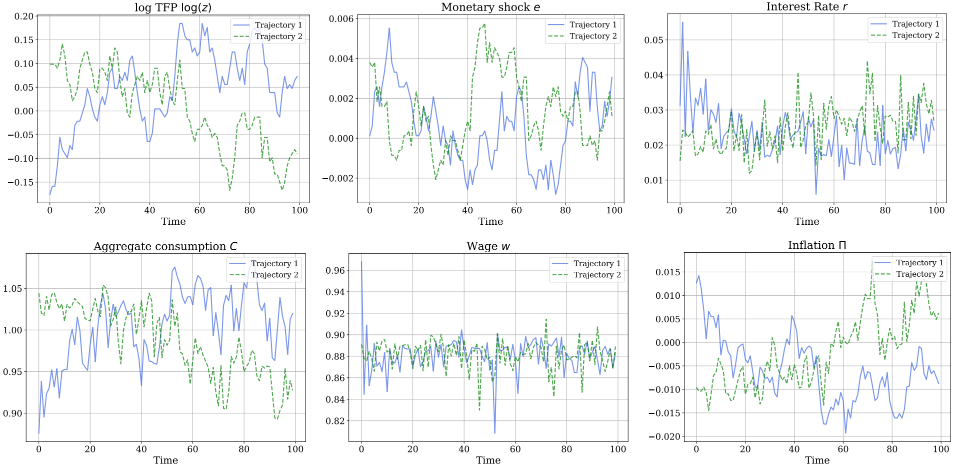
Symmetric treatment of firms and households, update policies simultaneously

In practice: good convergence properties

HANK: Household and firm policy functions



HANK simulations



Summary

Efficient and flexible global solution method for non-stationary HA models

"Structural RL": hybrid RL method that exploits known model structure

- RL about eqm prices but not individual states
- sidestep infinite-dimensional Master equation
- solve much lower-dimensional problem

Solves problems traditional methods struggle with

- non-trivial market clearing conditions
- HANK with forward-looking Phillips curve

Summary

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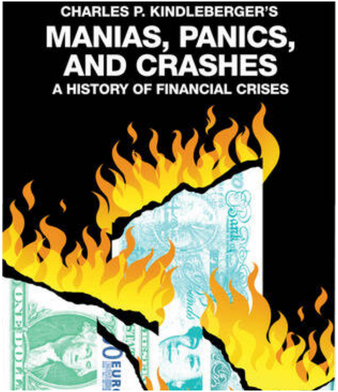


Solves problems traditional methods struggle with

- non-trivial market clearing conditions
- HANK with forward-looking Phillips curve
- next: models of large crises, booms/busts

Thanks!

In stationary world, lagged prices are enough for RE ▸ back

Recall Assumption 1: agents observe prices p_t but not distribution $G_t(s)$

Important: Assumption 1 still consistent with rational expectations

Why? Wold representation theorem!

Step 1 (Wold): if p_t-process is stationary, it has Wold representation = VMA(∞)

$$p_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}, \qquad c_j = \text{some unknown coefficients}$$

Step 2: if VMA(∞) is invertible, it can be expressed as a VAR(∞) and hence

$$p_{t+1} \sim \mathcal{T}_p(\cdot | p_t, p_{t-1}, \dots)$$

In practice, include finitely many lags

Assumption 2: extreme case with zero lags $p_{t+1} \sim \mathcal{T}_p(\cdot | p_t)$

Key difficulty: equilibrium prices are not Markov

- Equilibrium prices satisfy ▸ back

$$
\begin{aligned}
p_t &= P^*(\mathbf{g}_t, z_t) \\
\mathbf{g}_{t+1} &= \mathbf{A}^\top_{\pi_t(z_t)} \mathbf{g}_t \\
z_{t+1} &\sim \mathcal{T}_z(\cdot|z_t)
\end{aligned}
$$

- Difficulty: low-dimensional p_t does not have Markov property...
- ... only extremely high-dimensional $(\mathbf{g}_t, z_t)$ does
- Dynamic programming can only handle Markov states $\Rightarrow$ Master equation

$$V(s, \mathbf{g}, z) = \max_c u(c) + \beta \mathbb{E}\left[V(s', \mathbf{g}', z')|s, \mathbf{g}, z\right] \quad \text{s.t. } s' \sim \mathcal{T}_s(\cdot|s, c, P^*(\mathbf{g}, z))$$

- Without Markov transition prob's: cannot even write Bellman equation!
- But what if there was a way to approximate value and policy functions with p_t process for which there are no Markov transition probabilities?

Brief primer on reinforcement learning

RL: learning value & policy functions in incompletely-known Markov decision processes from experience (Monte Carlo sampling) a.k.a. “approximate DP”

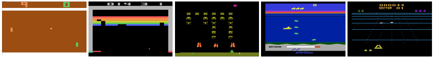
Playing Atari with Deep Reinforcement Learning

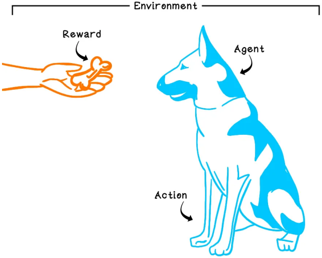
Volodymyr Mnih Koray Kavukcuoglu David Silver Alex Graves Ioannis Antonoglou

Daan Wierstra Martin Riedmiller

DeepMind Technologies

{vlad,koray,david,alex.graves,ioannis,daan,martin.riedmiller} @ deepmind.com





Computing an expected value

Random variable x

How compute expected value $\mathbb{E}[x]$? Two approaches:

1. Exact: know probability distribution $f(x) \Rightarrow$ calculate

$$\mathbb{E}[x] = \int x f(x) dx$$

2. Monte Carlo: don't know f but can sample $\{x_1, x_2, \ldots, x_N\}$

$$\mathbb{E}[x] \approx \bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

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$$\mathbb{E}[x] \approx \bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$
Or update incrementally (stochastic approximation method):
$$\bar{x}_k = \frac{1}{k}\sum_{i=1}^{k} x_i \quad \text{satisfies} \quad \bar{x}_k = \bar{x}_{k-1} + \frac{1}{k}\left(x_k - \bar{x}_{k-1}\right), \quad \frac{1}{k} = \text{“learning rate”}$$

Computing a value function

For now: eliminate actions and individual states

$$v_0 = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t u(p_t)\right], \quad p_t = \text{exogenous stochastic process}$$

Two approaches:

1. Dynamic programming: p_t Markov and know $f(p'|p)$

$$v(p) = u(p) + \int v(p')f(p'|p)dp'$$

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2. Monte Carlo: don't know f but sample N trajectories $\{p_t^i\}_{t=0}^{T}$

$$v_0 \approx \widehat{v}_0 = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \beta^t u(p_t^i) \quad \text{or} \quad \widehat{v}_0^k = \widehat{v}_0^{k-1} + \frac{1}{k}\left(\sum_{t=0}^{T} \beta^t u(p_t^k) - \widehat{v}_0^{k-1}\right)$$

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Can also be extended to compute optimal policy: policy gradient method

Computing a value function

For now: eliminate actions and individual states

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Two approaches:

1. Dynamic programming: p_t Markov and know $f(p'|p)$

$$v(p) = u(p) + \int v(p') f(p'|p) dp'$$

2. Temporal difference learning: N trajectories, update v incrementally

$$\widehat{v}^k(p_t^k) = \widehat{v}^{k-1}(p_t^k) + \frac{1}{k}\left(\left[\sum_{\tau=0}^{n-1} \beta^\tau u(p_{t+\tau}^k) + \beta^n \widehat{v}^{k-1}(p_{t+n}^k)\right] - \widehat{v}^{k-1}(p_t^k)\right)$$

Can also be extended to compute optimal policy: policy gradient method

Agent vs Environment, Rollout of a Policy

RL distinguishes between agent and environment = everything outside of agent

In HA macro:

- agents = households and their policies
- environment = everything else, including market clearing etc

Related: rollout = agent interacting with environment under given policy

- take any (generally suboptimal) policy, run it forward in time
- how optimize policy is completely separate question (policy improvement)

This viewpoint will be important, in particular for non-trivial market clearing

back